THE ECONOMICS OF EDUCATION

A COMPREHENSIVE OVERVIEW

SECOND EDITION

EDITED BY
STEVE BRADLEY AND COLIN GREEN

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Foreword

Economics is concerned with the efficient allocation of scarce resources in order to maximize productivity, output or more generally social outcomes. This is no less the case for the study of education, whether it be publically funded or privately funded. In addressing the overarching issue of making an efficient allocation of resources in education, economists and other academic disciplines, such as educationalists and sociologists have debated whether funding and provision should be separated. With this said, introducing market elements is becoming commonplace across the world and at all levels of education—primary, secondary and tertiary.

More specifically, in the previous edition of this volume, Brewer and McEwan (2010) highlight the development of the economic study of education from two initial analogies with economic production. First, the development of Human Capital Theory by Gary Becker and Jacob Mincer where individuals invest in their own education in a way that is analogous to physical capital, and in the same way, generates a stream of future returns. This initial work lead to a subsequent explosion in empirical work aimed at estimating rates of return to education across the world, and following that, a body of work aimed at honing-in on causal estimates of individual returns to schooling. At the same time, from the Coleman Report onwards, economists have endeavored to understand the determinants of test scores, focusing on how educational inputs (e.g., schools, families, policies) are transformed into educational outputs. Again, this has led to a voluminous body of research aimed at understanding what are the most critical educational inputs such as school expenditure, class size, teachers, and family inputs.

Since the publication of the first volume in 2010, the field of the economics of education has grown dramatically both in size and scope. The growth in empirical work in particular is notable. The study of education economics has been at heart of the so-called *identification revolution* and the current volume seeks to reflect this. While the book continues to cover the core issues of returns to education and determinants of test scores, there are additional sections aimed at reflecting developments in the behavioural economics of education and recent policy developments. The book is split into five parts.

Section 1: Methods and overview

The first two chapters seek to provide an overview of current empirical approaches in the economics of education (Schwerdt and Woessmann) and the key insights from behavioural approaches to the economics of education (Nielsen and Damgaard). These are important standalone chapters in their own rights, but also provide an overview of current approaches useful in reading many of the following chapters. They also highlight the development of econometric techniques to address the key questions in the economics of education.

Section 2: Private and social returns to education

Understanding the returns to education remains a core issue, both in developed economics (Gunderson and Oreopolous) and in developing economies (Patrinos and Psacharopolous). On a related theme, the rapid and ongoing expansion of higher education across a range of countries leads naturally to questions of the returns to a degree qualification (Walker). At the same time, there is a growing emphasis on the role of educational quality (Hoekstra). Similarly, the returns to education are wider than earnings alone, and cover areas such as children's health (Huebener), reduced crime (Lochner) and pro-civic behavior (Dee).

Insofar as the provision of, and investment in, schooling is not uniform across economics it has clear role in the development of societal inequality. Reflecting this, there are also clear roles for educational investment as determinants of intergenerational inequality (Blanden) and racial inequalities (Carnoy). A central aspect of educational decisions is the choice to quit schooling, and the consequences and social desirability of this decision (Rumberger).

Section 3: Production, costs and financing

How to distribute educational inputs and how these influence educational outputs is central to the development of efficient schooling and educational systems. An initial chapter provides an overview of the education production function approach to understanding this process (Hanushek), the relationship of educational attainment with economic growth (Woessmann and Hanushek), developing country evidence on the educational production (Glewwe, Lambert and Chen), how education is utilised in the labour market in developing countries (Battu and Bender), and the efficiency of educational production (Johnes).

The development of research on educational production has led to increased interest in specific inputs into the production function such as parental and family inputs (Currie and Goodman; Rabe), early childhood interventions (Nores), Class size (Schanzenbach), teacher quality (Strøm and Falch) and the role of peers (Paloyo). Recent research from the behavioural perspective highlights the role of both monetary and non-monetary incentives in the production of education (Wagner and Schildberg- Hörisch).

Finally, two chapters provide a discussion of educational financing. One in the form of an overview of issues in school financing (Rice), and another on tuition and fees (Ehrenberg).

Section 4: Teacher labour markets

Teachers provide the main labour input into the production of education and this naturally has led to a voluminous literature on teachers and teacher labour markets. This section starts with an overview of teacher labour markets (James and Wyckoff), with companion chapters that focus on the specifics of teachers in developing countries (Glewwe, Shen, Sun and Wisniewski) and issues of teacher labour supply (Dolton). Further chapters consider key issues such as recruitment and retention (Loeb), compensating wage differentials (Feng) and incentives for teachers (Santibanez).

Section 5: Education market, choice and incentives

The final section of the second edition, returns to the issue of the funding and provision of education, and examines the role of the state and the "market" in education. At the heart of much of this literature is the issue of to what extent the state funds and provides education (Plank and Davies). While a related question is whether and why firms provide vocational training, and

how this affects individual outcomes (Muehlemann and Wolter). More generally, a variety of different approaches to the broad issues of education provision and increasing efficiency have been tried, reflecting the different institutional settings in each country. Several themes have emerged in the literature, such as the role of parental choice vis-à-vis primary and secondary schools, and the impact, if any, of introducing competition between schools for pupils, whilst also addressing the question of school accountability, in an attempt to increase test scores (Figlio and Ladd; Bradley; Nechyba; Foliano and Silva). While a related question is what is the role and effect incentives aimed at improved educational performance (Eide and Showalter). The effects of private provision (Green), catholic school provision (Sander) and other models of provision such as charter schools (Kho) are also covered in this section.

Final comments

We hope that this volume introduces students, practitioners and policy makers to the subject matter of the economics of education, and does so in an accessible way for those who do not have any training in economics. We also hope that the volume builds on the first edition of this volume, whilst illustrating the rapidly developing nature of this field of academic endeavor.

Steve Bradley
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CHAPTER

1

Empirical methods in the economics of education

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Introduction

Empirical research in the economics of education often addresses causal questions. Does an educational policy or practice cause students' test scores to improve? Does more schooling lead to higher earnings? This article surveys the methods that economists have increasingly used over the past two decades to distinguish accidental association from causation.[1] The methods include research designs that exploit explicit randomization as well as quasi-experimental identification strategies based on observational data. All methods are illustrated with a range of selected example applications from the economics of education.

From correlation to causation

It is reasonably straightforward to establish whether there is an association between two variables using standard statistical methods. Understanding whether such a statistical correlation can be interpreted as a causal effect of one variable, the *treatment*, on the other, the *outcome*, is, however, another question. The problem is that there may well be other reasons why this association comes about. One reason would be "reverse causality", which describes a situation where the outcome of interest asserts a causal effect on the treatment of interest. Another example of alternative reasons for the association is that of "omitted variables", where a third variable affects both treatment and outcome.

Whenever other reasons exist that give rise to a correlation between a treatment and an outcome, the overall correlation cannot be interpreted as a causal effect. This situation is commonly referred to as the *endogeneity problem*. The term originates from the idea that the treatment cannot be viewed as exogenous to the model determining the outcome, as it should be, but that it is rather endogenously determined within the model—depending on the outcome or being jointly determined with the outcome by a

[1] The exposition here is partly based on Schlotter, Schwerdt, and Woessmann (2011).

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third factor. Because of the problem of endogeneity, simple estimates of the association between treatment and outcome based on correlations will be biased estimates of the causal effect of treatment on outcome.[2]

Standard approaches such as multivariate regression models try to deal with this problem by observing other sources of a possible correlation and by taking out the difference in outcomes that can be attributed to these other observed differences. This allows estimating the association between treatment and outcome conditional on the effects of observed factors. The required assumption for the identification of a causal effect in this case is often called *selection-on-observables* assumption. It implies that the conditional estimate identifies the causal effect of interest if selection into treatment is sufficiently described by the observed variables included in the model. However, more often than not, one cannot observe all relevant and non-ignorable variables. But as long as part of the omitted variables stay unobserved, the estimated conditional association will not necessarily warrant a causal interpretation.

Over the past two decades, it has become increasingly apparent in the literature on the economics of education that there are myriad important factors that remain unobserved in our models of interest, often rendering the attempts to control for all relevant confounding factors in vain. Just think of such factors as innate ability of students, parental preferences for certain outcomes, the teaching aptitude of teachers, or the norms and values of peers and neighborhoods. Even if one manages to obtain observable measures of certain dimensions of these factors, others—often important ones—will remain unobserved. Even more, controlling for observable factors does not solve the endogeneity problem when it is due to plain reverse causality, in that the outcome causes the treatment. The only solution is to search for variation in treatment that is not related with other factors that are correlated with the outcome.

The same caveats that apply to the traditional models also apply to other techniques that ultimately rely on a selection-on-observable assumption such as propensity score matching. The central idea of this technique is to find matching pairs of treated and untreated individuals who are as similar as possible in terms of observed (pre-treatment) characteristics. Under certain assumptions, this method can reduce the bias of the treatment effect. But as long as relevant factors remain unobserved, it cannot eliminate the bias (see, e.g., Becker and Ichino (2002)). In this sense, matching techniques cannot solve the endogeneity problem and suffer as much from bias due to unobserved factors as traditional models.[3]

In this chapter, we turn to techniques, increasingly applied by economists, that aim to provide more convincing identification of causal effects in the face of unobservable confounding factors. In medical trials, only some patients get treated, and the assignment to the group of treated and non-treated patients is done in a randomized way to ensure that it is not confounded with other factors. The non-treated patients constitute a so-called *control group* to which the treated patients are compared. The aim of the discussed techniques is to mimic this type of experimental

[2] Other sources of endogeneity can be self-selection (objects with different characteristics can choose whether to be treated or not) and simultaneity (treatment and outcome are choice variables that are jointly determined). Measurement error in the treatment variable gives rise to a particular form of association between treatment and outcome (one that in the case of classical measurement error biases the estimates toward finding no effect).

[3] Matching techniques can still improve on the formation of proper treatment and control groups when they are combined with one of the approaches discussed below, where the latter ensures exogeneity of treatment and thus causal interpretation.

design, often using data not generated by an explicitly experimental design. The techniques aim to form a treatment group (that is subject to the treatment) and a control group (that is not subject to the treatment) which are exactly the same. That is, they should not have been sub-divided into treatment and control group based on reasons that are correlated with the outcome of interest. Ideally, one would like to observe the same individuals at the same point in time both in the treated status and in the non-treated status. Of course, this is impossible, because the same individual cannot be in and out of treatment at once. Therefore, the key issue is estimating what would have happened in the *counterfactual*—which outcome a treated individual would have had if she had not been treated.

The central idea of these techniques is that if the separation of the population into treatment and control group is purely random and a sufficiently large number of individuals is observed, then randomness ensures that the two groups do not differ systematically on other dimensions. In effect, the mathematical law of large numbers makes sure that the characteristics of those in the treatment group will be the same as those in the control group. Thus, the causal effect of the treatment on the outcome can be directly observed by comparing the average outcomes of the treatment group and the control group, because the two groups differ only in terms of the treatment. The aim of the empirical methods discussed in this chapter is to generate such proper treatment and control groups and thus rule out that estimates of the treatment effect are biased by unobserved differences.

Explicit randomization

We start our discussion with two techniques that use explicit randomization to construct valid treatment and control groups that, on average, do not differ from one another in observed or unobserved dimensions.

Randomized controlled experiments

From a conceptual perspective, randomized controlled experiments—or randomized controlled trials (RCTs)—constitute an important benchmark with certain ideal features against which to judge the other techniques discussed in this chapter. Because of their desirable features, controlled experiments are sometimes being referred to as the most rigorous of all research designs or even as "gold standard" (e.g., Angrist (2004)). The most important feature is the random assignment of participants to treatment and control groups.

Suppose we are interested in evaluating whether a specific educational program has a causal effect on economic outcomes of individuals who participated in the intervention. Ideally, we would like to observe the same individuals in a counterfactual world and compare the two individuals' outcomes. As this is impossible, it is necessary to find research designs that provide good estimates of counterfactual outcomes. To this end, researchers typically try to build two groups that are "equivalent" to each other. One group (the treatment group) is assigned to the specific program, the other group (the control group) is not. As the groups consist of different individuals, creating two groups that are completely equal regarding all important aspects such as family background, social status, and the like is not possible. Thus, it becomes rather a question of probability in the sense that the task is to create two groups which should be on average equal with sufficiently high probability. This is possible by randomly drawing a sufficiently large number of individuals from a population and randomly assigning them to the treatment and the control group. Random assignment is of crucial importance in this context, because if the two groups are indeed on average identical apart from the assignment to the program, all observed differences in outcomes between the groups can be attributed to the average effect of the program.

A classic example of a controlled experiment is the Perry Preschool Program, which nicely illustrates the advantages, but also some limitations and caveats, of explicit experiments. The experiment was conducted in 1962 in Ypsilanti, Michigan, when school authorities recognized the low school performance of at-risk children from poor neighborhoods compared to better-off children. In order to improve the social and cognitive development of the children, they decided to test whether an intense intervention of high-quality early childhood education could help these children (see Barnett (1985)). The crucial feature of this program is that it was conducted as a controlled experiment. In particular, 123 at-risk children at ages 3–4 were randomly assigned to a treatment and a control group (see Belfield, Nores, Barnett, and Schweinhart (2006)). The 58 children in the treatment group received a high-quality preschool program for one or two (academic) years, including daily center-based care in small groups, home visiting each weekday, and group meetings of the parents. The 65 children in the control group did not receive these services. The members of both groups were followed until adulthood and have been surveyed several times. Ultimately, the study can document positive effects of the two-year preschool program on desirable outcomes in several dimensions. Among others, at the age of 40 the treated individuals had significantly higher earnings, higher probability of high-school completion, and fewer incidents of crime than the adults from the control group.

Research on several other topics in education has made use of RCTs. Another of the most well-known is the so-called Project STAR in the US state of Tennessee that randomly assigned students and teachers to classes of different size. This class-size experiment has been subjected to extensive research (e.g., Finn and Achilles (1990)) that, among others, also takes into account incomplete randomization caused by later switching between classes due to behavioral problems of students and relocation of families (see Krueger (1999)).[4] Recent studies using the Project STAR experiment have investigated long-term impacts of classrooms and peers (Chetty et al., 2011; Bietenbeck, 2019).

RCTs are also increasingly being used to answer various other questions in the economics of education (see Fryer (2017) for a survey). For example, several RCTs have been implemented to study the effects of incentives for students on educational outcomes. Fryer (2011), Bettinger (2012), and Levitt, List, Neckermann, and Sadoff (2016) conduct a series of randomized field experiments on financial incentives and student achievement. The randomized treatments in separate experiments include financial rewards for performance on a test, for grades in core courses, for performance on a metric that includes attendance, behavior, and inputs to educational production functions chosen by schools, as well as offering students money to read books. In higher education, Bettinger, Long, Oreopoulos, and Sanbonmatsu (2012) implement an RCT to estimate the effect of application assistance and information on college decisions. Also on the classic question in the economics of education of how education affects labor-market outcomes, randomized controlled experiments have been implemented by sending out CVs with randomized education elements for application (e.g., Deming, Yuchtman, Abulafi, Goldin, and Katz (2016); Piopiunik, Schwerdt, Simon, and Woessmann (2018)).

Today, RCTs in schools are also extensively used in the context of developing countries. For example, Muralidharan and Sundararaman (2011) and De Ree, Muralidharan, Pradhan,

[4] However, substantial concerns remain with the implementation of the experiment which may compromise its evidence on the effects of class size on student achievement (see Hanushek (1999)).

and Rogers (2018) implement experiments on teacher pay in India and Indonesia.

Randomized controlled experiments are, however, not free of potential caveats and drawbacks. One important drawback of explicit experiments is that participants are often aware of their participation in an experiment.[5] Thus, they may change their behavior exactly because they know that they are observed. This so-called *Hawthorne effect* is an example of how evidence drawn from controlled experiments can be compromised. This should be especially the case if one result of the experiment appears more favorable to participants than the other result.

Another concern is that it is often not clear to which extent the results of an experiment can be generalized. In case of the Perry Preschool Program, for example, the choice of the underlying sample was explicitly targeted at at-risk children, limiting the "*external validity*" of the study to this sample. That is, the causal inference is limited to the specific group of African-American children from disadvantaged backgrounds. If the selection process had been completely random, effects might have differed. But generalized results that are valid for the whole population were not the aim of this specific experiment. Focusing on the effects for specific sub-groups is often more interesting and relevant from a policy point of view.

External validity may also be hampered by the fact that a full-scale implementation of a policy could generate general-equilibrium effects. If, for example, a small-scale experiment leads to more schooling and, therefore, increased earnings among treated participants, a full-scale intervention might not generate the same effects on earnings because a substantial increase in the supply of highly-educated workers may decrease the marginal returns to schooling in general equilibrium. In addition, there can be several factors that complicate both the random draw from the population and the random assignment to the groups. For example, Heckman, Moon, Pinto, Savelyev, and Adam (2010) show that the Perry Preschool Program did not fully reach random assignment, as treatment and control group individuals were reassigned afterward. More generally, controlled experiments often suffer from issues of non-perfect implementation, compromising the validity of their results.

Lotteries of oversubscribed programs

Researchers in the economics of education often use another form of explicitly controlled experiments as the basis of their empirical strategy. Sometimes, when an institution aims to implement a specific educational intervention, resources are not enough to finance participation for everybody who is interested. In such a setting, the assignment of an oversubscribed program is often handled by randomized lotteries, so that each applicant has the same chance of participation. Among those who apply for the program, this then boils down to being an explicitly randomized set-up. Lottery winners constitute the treatment group in this case, while lottery losers form the control group. For an empirical evaluation of the program, researchers then only need to collect data on outcomes and other characteristics of the two groups, preferably also before but in particular after implementation of the program.

A well-known example of randomized lotteries of oversubscribed programs is that of charter schools in the US. Compared to traditional public schools, charter schools are granted

[5] Ideally, participants should not be aware of the fact that they are part of an experiment. In their experiment analyzing the use of randomly distributed vouchers for adult training courses, Schwerdt, Messer, Woessmann, and Wolter (2012) created a setting in which both groups were completely unaware that they were investigated.

more autonomy to staff their own classrooms, choose their own curricula and manage their own budgets. Often, demand for seats in particular charter schools is so high that they must admit students by random lottery. This feature of the admission process has been exploited by several studies to investigate whether charter school attendance has a positive effect on student achievement (see Epple, Romano, and Zimmer (2016) and Chabrier, Cohodes, and Oreopoulos (2016) for recent surveys). The treatment group in these studies are students who won a place in a charter school, and the control group are those who did not win a place. Access to administrative data on student achievement then allows for a comparison of later outcomes of the two groups. This set-up allows to estimate both the effects of the offer of a place in a charter school (the "intention-to-treat" effect) and the effects of actually attending a charter school (the "treatment-on-the-treated" effect). Abdulkadiroğlu, Angrist, Dynarski, Kane, and Pathak (2011), for example, analyze student achievement in Boston's traditional public schools as well as in charter schools. Exploiting randomized admission lotteries, they find large positive effects of charter schools in this setting.

In a similar spirit to charter school evaluations, Deming, Hastings, Kane, and Staiger (2014) study whether school choice has an impact on postsecondary attainment exploiting a public school choice lottery in Charlotte-Mecklenburg. Angrist, Bettinger, and Kremer (2006) estimate the effects of school vouchers in Colombia exploiting a randomized lottery. A similar approach is also employed by Cullen, Jacob, and Levitt (2006) to estimate the effects of increased choice among specific public schools in Chicago. Bettinger and Slonim (2006) perform a laboratory experiment within the field setting of a voucher lottery in Toledo, Ohio, to estimate the effect of school vouchers on altruism.

An obvious key advantage of studies based on randomized lotteries of oversubscribed programs is that they do not require setting up a separate experiment, but rather build on the randomization that is implemented anyway. In addition, these programs tend to refer to field trials that are enacted in a real-world setting, rather than an artificial experimental setting. Similar to explicit experiments, evaluations of randomized lotteries of oversubscribed programs may be subject to Hawthorne effects and may miss general-equilibrium effects. In addition, motivating those who lost in the lottery to participate in subsequent surveys and tests may not be easy.

A key shortcoming of oversubscribed lotteries is that the very fact that specific schools or programs are oversubscribed may imply that their productivity is higher than that of other schools or programs that do not attract as many students. If parents and students vote with their feet, oversubscription status indicates preferred schools, so that average effects would be expected to be smaller than effects of oversubscribed schools. In fact, Abdulkadiroğlu et al. (2011) provide suggestive evidence that the effect of Boston charter schools that are not oversubscribed may be substantially smaller than the effect of the oversubscribed charter schools, thus limiting the external validity of results for oversubscribed schools.

Another potential drawback of the setting of randomized lotteries of oversubscribed programs is that the underlying population is just those individuals who applied for participation in the program. This will not necessarily be a random draw from the population at large. In particular, subjects who particularly like the program, who view a particular need for the intervention, or who place particular value on the outcome of the intervention may be more inclined to apply than the average population. In the context of charter school evaluations, this implies that lottery-based studies do not necessarily generalize to the vast number of students who do not wind up in admission lotteries. However, recent comparisons of results from other research designs that do not exploit

admission lotteries find comparable results (e.g., Abdulkadiroğlu et al. (2011); Abdulkadiroğlu, Angrist, Hull, and Pathak (2016)).

Natural experiments

In the absence of intentional randomization, identifying causal effects is a challenging task. However, sometimes it is possible to exploit variation in observational data that stems from sources that are exogenous to the association of interest. In particular, two techniques try to mimic the random assignment of controlled experiments by building on incidents where nature or institutional rules give rise to exogenous variation. Such identification strategies are also referred to as *natural experiments* or *quasi-experiments*.

Instrumental-variable approach

The instrumental-variable (IV) approach is an identification strategy that tries to get close to the set-up of a controlled experiment using observational data. It aims to identify variation in the exposure to a certain education policy or practice that stems from a particular source that is not correlated with the outcome of interest.

The idea behind this approach is simple. Think of the treatment of interest as having two parts. One part is subject to the endogeneity problems discussed above. The other part does not suffer from endogeneity problems and can thus be used for causal identification. The IV approach aims to isolate the latter part of the variation in the treatment variable. This is achieved by using only that part of the variation in the treatment variable that can be attributed to an observed third variable (the *instrument*) which is not otherwise correlated with the outcome (or with omitted variables that are correlated with the outcome). Having information on such an instrument allows to isolate variation in treatment that is exogenous to the model of interest and thus to obtain unbiased estimates of the causal effect of treatment on outcome.

The key to success of any IV approach is to find a convincing instrumental variable—one that is strongly associated with the treatment variable ("*instrument relevance*"), but is not correlated with the outcome apart from the possible indirect effect running through treatment ("*instrument exogeneity*"). If such an instrument can be found, it is possible to identify the treatment effect through a part of the variation in treatment that is triggered by variation in the instrumental variable, thereby overcoming problems such as reverse causality and omitted variables and achieving consistent estimation.

Changes in compulsory schooling laws constitute a good example that illustrates the use of an IV identification strategy in educational research. Several studies exploit arguably exogenous changes in the educational attainment of individuals caused by changes of the minimum school-leaving age as an instrument for actual education attainment (e.g., Oreopoulos (2006); Black, Devereux, and Salvanes (2008); Cygan-Rehm and Maeder (2013); Piopiunik (2014b); Fort, Schneeweis, and Winter-Ebmer (2016); Hanushek, Schwerdt, Wiederhold, and Woessmann (2015)). As a seminal study, Harmon and Walker (1995) estimate the returns to schooling on the labor market in the United Kingdom. To cope with the endogeneity of educational attainment, they use variables indicating changes in laws determining the minimum school-leaving age as an instrumental variable for years of schooling. This is possible as individuals in the sample (employed males aged 18–64) faced different minimum school-leaving ages during their youth because two legislative changes raised the minimum school-leaving age from 14 to 15 in 1947 and from 15 to 16 in 1971. The setup aims to meet the two key assumptions of IV. First, an increase in the minimum school-leaving age induces at least part of the population to stay in school longer. Second, this change in legislation should have no effect on

individuals' earnings other than the indirect effect through increased schooling. The IV estimates of Harmon and Walker (1995) suggest that an additional year of schooling raises earnings by 15%—roughly three times the corresponding OLS estimate.[6]

However, IV estimates should always be interpreted carefully. Angrist, Imbens, and Rubin (1996) show that IV procedures estimate the effect of schooling only for that subgroup of the population that complies with the assignment, i.e., that actually changes the schooling decision because of a change in the instrument. Thus, one has to interpret the IV estimate as a so-called Local Average Treatment Effect (LATE), i.e., as applying only to the "local" sub-population that is affected by the instrument. In the example above, this suggests that the estimates reflect returns to schooling for those individuals with minimum schooling, whose schooling decisions are affected by the laws on minimum school-leaving ages. Thus, effects identified by IV estimation do not necessarily reflect average effects for the entire population, raising points of external validity in the same way as controlled experiments.

Several other studies in education rely on instruments for the identification of causal effects. Currie and Moretti (2003) use the availability of colleges in a woman's county in her 17th year as an instrument for maternal education in the United States and find that higher maternal education improves infant health. Machin and McNally (2008) exploit changes in the rules governing the funding of information and communication technology (ICT) across English school districts in an IV specification to estimate the effect of ICT spending on student performance. West and Woessmann (2010) exploit the historical pattern that countries with larger shares of Catholics in 1900 tend to have larger shares of privately operated schools even today. Using historical Catholic shares as an instrument for contemporary private competition, they investigate the effect of private competition on student achievement in a cross-country setting. Jackson, Johnson, and Persico (2016) study the effects of school spending on educational and economic outcomes and use the timing of the passage of court-mandated reforms and their associated type of funding formula change as source of variation in school spending in the United States.

These examples illustrate that IV approaches exploit arguably exogenous variation from very different sources. In practice, the "trick" of any IV approach is to find a good instrument. If a convincing instrument is found, causal effects can be well identified even with purely cross-sectional observational data. However, the main assumptions of IV—instrument relevance and exogeneity—must be carefully evaluated in any application.

A key advantage of such quasi-experimental analyses is that they circumvent some of the leading problems with RCTs, such as the facts that they are expensive, time-consuming, and difficult to explain to public officials whose cooperation is generally needed. In addition, quasi-experimental studies are not subject to Hawthorne effects because subjects are not aware that they are part of an experiment, and well-designed natural experiments can be able to capture general-equilibrium effects that RCTs usually cannot.

Regression-discontinuity approach

Another approach in the spirit of natural experiments is the regression-discontinuity (RD) approach. This approach is typically used in a specific setting where a treatment is determined by whether a subject falls above or below a

[6] Note, however, that Stephens and Yang (2014) find that similar results for the United States are not robust to allowing for differential year-of-birth effects across the four US census regions

certain cutoff value of a specified *assignment variable* (also called *running* or *forcing variable*).

The study by Papay, Murnane, and Willett (2016) nicely illustrates the use of the RD approach. They study whether performance labels in secondary school affect the college-going decisions of students. This is a challenging research question, because students typically receive good or bad performance labels for a reason. Thus, based on observational data it is difficult to disentangle the isolated effect of receiving a specific performance label from any effects that the underlying determinants of performance may have on future outcomes. Studying this question in the context of a controlled experiment is also unlikely to be an option, because it would be unethical to randomly assign actual performance labels to students for the purpose of an experiment.

Here RD can help. In the setting of Papay et al. (2016), students participating in state-mandated standardized tests receive a score between 200 and 280 as well as a label that summarizes their performance on these tests. In particular, students scoring more than 260 points on the tenth-grade mathematics test are labeled "advanced". The idea of the RD design then is to compare students in a sufficiently small range just above and below the cutoff value of 260, where those above form the treatment group and those below constitute the control group. Conditional on any continuous effects of the test score, students just above and just below the cutoff will in expectation not differ by more than the treatment (the label), because they are very similar in terms of the assignment variable (the test score). The comparison of units that are in a sufficiently small range below and above the threshold therefore comes close to an experimental setting with random assignment to treatment and control groups. Any jump or discontinuity in the outcome (the probability of attending college) that can be observed at the threshold can then be interpreted as the causal effect of receiving the performance label. In addition, the fact that the assignment to the treatment and control groups follows a non-linear pattern—the discontinuity at exactly the cutoff value—allows the RD approach to control for any smooth function of the variable determining eligibility. The assumption required for the RD approach to capture the causal effect (the identifying assumption) thus is that there are no other discontinuities around the cutoff.

A nice feature of the RD design is that is has a straightforward graphical depiction. When plotting the outcome of interest against the assignment variable, there should be a clear jump in the outcome at the assignment threshold that determines the treatment status of students (see Fig. 1.1). In the study by Papay et al. (2016), there is a jump in the probability of attending college at the threshold. To show that this jump is really caused by the performance label, they estimate local linear regression models controlling for test scores using only observations that fall within a small bandwidth (8 points in their preferred specification) on either side of the cutoff. Their results corroborate the graphical finding that receiving the "advanced" label on the tenth-grade mathematics test increases the probability that urban, low-income students enroll in college.

In similar setups, several recent studies apply RD designs to estimate the causal effects of grade retention on future outcomes (Jacob and Lefgren (2004; 2009); Eren, Depew, and Barnes (2017); Schwerdt, West, and Winters (2017)). Many US states have recently enacted policies requiring that students who do not meet minimum performance expectations on state-mandated standardized tests at the end of specific grades be retained and provided with remedial services. In this setting, scoring just below a threshold level induces a specific treatment. However, due to the availability of exemptions for students scoring below the promotion cutoff and voluntary retention of some higher-scoring students, the treatment in this case is not assigned deterministically by the assignment variable. Thus,

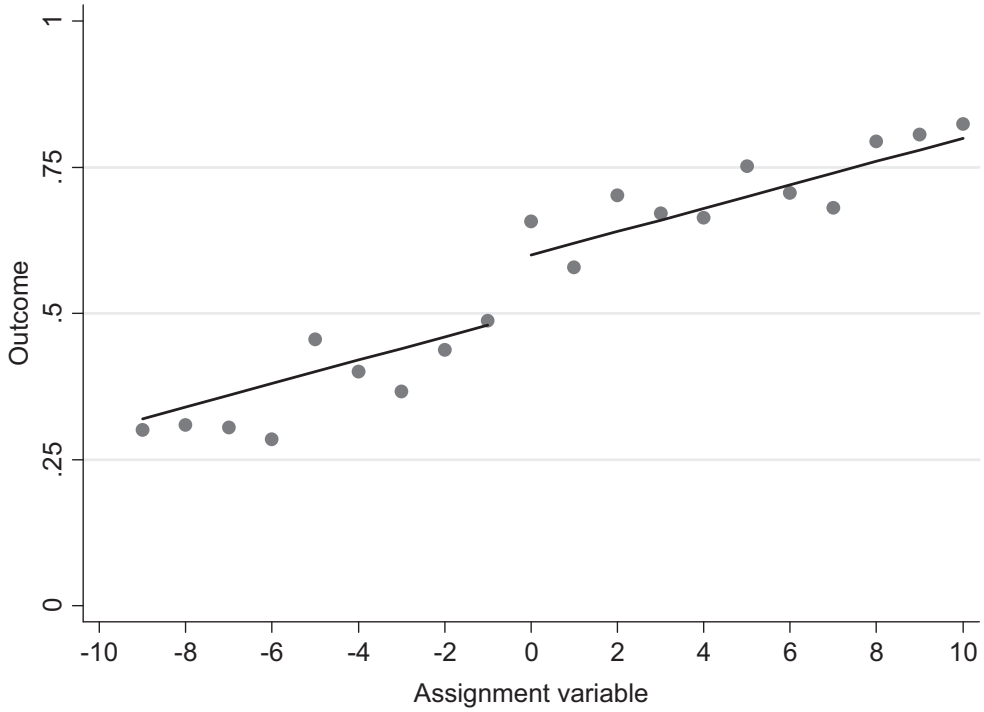

FIG. 1.1 Stylized exposition of the RD design.

in this case we rather expect to see a jump in the probability that students repeat the grade at the assignment threshold. In contrast to so-called sharp RD designs where the cutoff unequivocally divides observations into a treatment and a control group, such settings lead to so-called "fuzzy" RD designs. These designs exploit discontinuities in the probability of treatment and can be implemented as an instrumental-variable approach where the discontinuity acts as the instrument for treatment status.

Several other examples illustrate that there is a rich number of cases where educational policies and practices are implemented in a manner that involves a discontinuity which allows for evaluation through the RD approach. Prominent examples are studies exploiting discontinuities in average class size due to maximum class-size rules to study the effects of class size on various outcomes (e.g., Angrist and Lavy (1999); Woessmann and West (2006); Fredriksson, Öckert, and Oosterbeek (2013); Leuven and Løkken (2018); Angrist, Lavy, Leder-Luis, and Shany (2018)). With maximum class-size rules, class sizes drop discontinuously whenever grade enrollment would lead to class sizes that exceed the maximum size determined by a specific rule. Other applications of the RD design use specified school-entry cutoff dates that lead to the effect that school entry ages vary due to the month of birth of the children (e.g., Bedard and Dhuey (2006); Mühlenweg and Puhani (2010); McCrary and Royer (2011)). Garibaldi, Giavazzi, Ichino, and Rettore (2012) study the effect of tuition fees on the probability of late graduation from university, exploiting the fact that tuition fees at Bocconi University in Milan are subject to discontinuous changes with respect to family income. Lavy (2010) uses a geographical RD approach that compares students in Tel Aviv

that enacted free school choice to students in neighboring areas that were not subject to the treatment. Abdulkadiroğlu, Angrist, and Pathak (2014) exploit admission discontinuities to estimate the effect of elite schools on students' achievement. Kirkeboen, Leuven and Mogstad (2016) use cutoffs in the admission to different fields and locations of higher education in Norway to estimate the effect of field of study on earnings.

The most serious threat to identification in the context of RD designs is manipulation of the assignment variable around the cutoff. Individuals often have incentives to receive the treatment, or not. Thus, they might try to manipulate the assignment variable. Think, for example, of teachers who manipulate results on standardized tests to ensure that specific students get promoted to the next grade. Such behavior would give rise to selection bias and violate the core assumption of no sorting around the cutoff. A common robustness check in any RD study is, therefore, to plot all observable exogenous characteristics of observational units against the assignment variable. Ideally, this exercise reveals no jumps for a long list of relevant observable characteristics at the cutoff, which would strengthen the case that there are also no jumps in unobservables at the cutoff, supporting the core assumption of the RD approach. A related robustness check is to check for discontinuities in the density of the assignment variable at the cutoff (see McCrary (2008)). Ideally, inspecting the distribution of the assignment variable reveals no suspicious clustering of observational units on either side of the cutoff.

Other issues with implementing the RD design are the choice of bandwidth and of the functional relationship with the assignment variable included in the empirical model. As a best-practice way of dealing with these issues, researchers typically report results for a variety of bandwidths and functional forms. Finally, due to the local identification around the threshold, external validity is also an issue for estimates based on the RD approach.

Methods using panel data

The availability of panel data allows for the application of two methods that do not rely entirely on strong selection-on-observables assumptions. Panel datasets are characterized by the fact that observational units are observed at least twice. The methods described here can be applied as long as some observational units change their treatment status between two incidents of observation. Incidents usually refer to two points in time, but they may also refer to two other dimensions such as different grade levels or subjects. The two approaches attempt to implicitly control for unobserved variables that would bias regression estimates of the causal effect of interest based on cross-sectional data.

Difference-in-differences approach

Difference-in-differences (DiD) approaches are applied in situations when certain groups are exposed to a treatment and others are not. The logic of DiD is best explained with an example based on two groups and two periods. In the first period, none of the groups is exposed to treatment. In the second period, only one of the groups gets exposed to treatment, but not the other. To provide an illustration, suppose that there are two classes in a given school observed at the beginning and the end of a school year. During this school year, only students in one of these two classes have additional afternoon lessons. DiD estimation can then be used to estimate the effect of additional lessons in the afternoon on student achievement.

The DiD is implemented by taking two differences between group means in a specific way (illustrated in Fig. 1.2). The first difference is the difference in the mean of the outcome variable between the two periods for each of the groups. In the hypothetical example, the first difference simply corresponds to the change in average test scores for each group between the beginning and the end of the school year. The second difference is the difference between

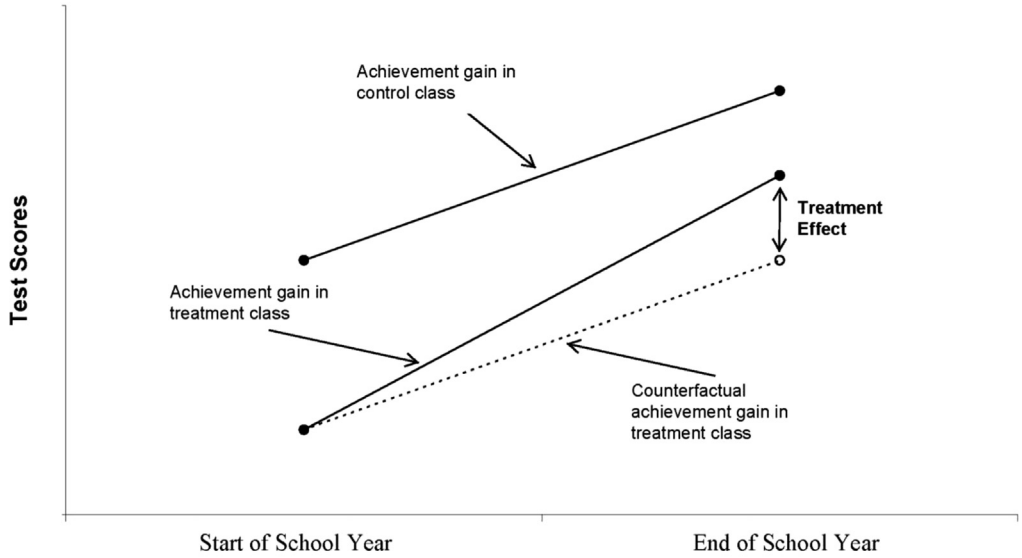

FIG. 1.2 Stylized exposition of identification in the DiD model.

the differences calculated for the two groups in the first stage (which is why the DiD method is sometimes also labeled "double differencing" strategy). This second difference measures how the change in outcome differs between the two groups, which is interpreted as the causal effect of the causing variable. Hence, in our example, the effect of afternoon lessons on student learning is identified by comparing the gains in average test scores over the school year between the two classes.

The idea behind the DiD identification strategy is simple. The two groups might be observationally different. That is, the group-specific means might differ in the absence of treatment. However, as long as this difference is constant over time (in the absence of treatment), it can be differenced out by deducting group-specific means of the outcome of interest. The remaining difference between these group-specific differences must then reflect the causal effect of interest.

The identification assumption of the DiD approach is that the group-specific trends in the outcome of interest would be identical in the absence of treatment. In terms of the hypothetical example, the identifying assumption is that both classes would have experienced the same increase in test scores over the school year in the absence of afternoon lessons. The assumption that the treatment class would have experienced a counterfactual achievement gain identical to the observed achievement gain in the control class is illustrated by the dotted line in Fig. 1.2. The plausibility of this identifying assumption depends on the specific setting to which DiD estimation is applied. If possible, researchers show that outcomes in the treatment and the control group prior to the treatment moved in parallel, which supports the assumption of parallel trends over the introduction of the treatment. In any case, the identifying assumption of the DiD approach is less restrictive than the assumption implicitly made in standard traditional methods, namely that the two groups are identical in terms of all relevant unobserved factors.

The DiD approach is particularly well-suited to estimate the causal effect of sharp changes in education policies or practices, providing

policy-makers with vital information even in the absence of controlled or natural experiments. It has, therefore, been used extensively to study the impacts of various education reforms around the world. Examples include reforms of compulsory schooling and tracking (e.g., Meghir and Palme (2005), Pekkala Kerr, Pekkarinen, and Uusitalo (2013), Meghir, Palme, and Simeonova (2018)), education priority zones for disadvantaged schools (e.g., Bénabou, Kramarz, and Prost (2009)), subsidized child care (e.g., Havnes and Mogstad (2011)), and paid parental leave (e.g., Danzer and Lavy (2018)).

DiD estimation of the effects of education reforms is feasible if the entire sample population is not exposed to the reform at the same point in time. This could be the case, for example, because of a gradual implementation of the new system across a country or because the reform is targeted only at a specific part of the population or at specific regions. For example, Piopiunik (2014a) evaluates a school reform in the German federal state of Bavaria that moved the timing of tracking in low- and middle-track schools from grade 6 to grade 4, while students in high-track schools were not affected. The DiD identification in this case rests on the comparison of the change in average outcomes between pre- and post-reform of students in high-track schools with the change in average outcomes over the reform of students in low- or middle-track schools. A potential violation of this assumption would, for example, be the presence of a general trend in Germany in the selection of students attending high-track schools. To eliminate such Germany-wide trends, performance is furthermore compared between students in Bavaria and students in other German states, where the timing of tracking did not change, which effectively extends the approach to a difference-in-difference-in-differences ("triple diffs") approach.

However, a DiD strategy does not always have to include a time dimension. For example, Hanushek and Woessmann (2006) investigate whether educational tracking affects school performance and inequality in the school system. Their research design is based on comparing outcome differences between primary school (when no country uses tracking) and secondary school (when some countries use tracking and others do not) in countries with and without tracked school systems. Moreover, the usefulness of DiD approaches also extends to settings other than sharp changes in education policies or practices. For example, Hanushek, Schwerdt, Woessmann, and Zhang (2017) employ a difference-in-differences approach that compares employment rates across different ages for people with general and vocational education to test whether gains in youth employment associated with a vocational education may be offset by less adaptability and diminished employment later in life.

Fixed-effects approach

In recent years, extensive datasets have become available, in particular in many US states, which provide performance data for every student in the public school system and allow following each student over several years. Such extensive panel datasets allow an even more extensive treatment of cases where unobserved differences are constant over time than traditional DiD approaches. These more extensive panel-data techniques can be applied to the case where unobserved factors are heterogeneous across individuals (rather than groups).

Presumably the most prominent factor in educational research that is usually not observed by researchers is the innate ability of individuals. In many empirical applications, a possible "ability bias" is a typical challenge. Think of research investigating whether the attendance of private schools improves students' achievements. In this case, estimates are only reliable if one can exclude that they are driven by differences in innate ability between students that choose to attend private and public schools. In the absence

of a clear measure for this factor and of a controlled or natural experiment, researchers sometimes try to circumvent this drawback by observing adequate proxy variables such as family background, parental education, and the like. This practical approach may help reducing the bias due to unobserved individual heterogeneity to some extent, but in most applications it presumably does not solve the problem completely.

While taking out fixed effects is also no panacea, it offers a solution to the problem in specific cases. As long as the factor that leads to unobserved heterogeneity—such as individual ability—has a constant impact on the outcome over time, it can be fixed by ignoring any variation in the level of outcomes across individuals and only focusing on changes over time. It is for this reason that panel data are needed that allow to observe the same individual at several points in time. In such a setting, the estimation can control for fixed effects of each individual in that it accounts for an indicator variable that takes out mean differences between individuals, so that only changes over time in inputs and outcomes are used to identify the effects of interest. This way, the estimation is able to control for unobserved but fixed heterogeneity across units of observation. In the most elaborate settings, such fixed effects can be introduced at the level of students, teachers, and schools.

An important special case of such models in the economics of education is the classical value-added approach that controls for previous outcomes of the same unit when estimating the effect of some input or intervention on current outcomes. In the special case of observing the same outcome (e.g., a test score) for a unit at two points in time, this approach estimates the effect of a specific education policy conditional on the previous outcome. This value-added approach is only a special case of more extensive models that use cross-sectional information of the same units (individuals, classes, schools) at several points in time in order to take out the effects of time-invariant factors.

Value-added models of educational production are most prominently used in the large literature investigating the contribution of teachers to educational achievements of students (see Hanushek and Rivkin (2012), Jackson, Rockoff, and Staiger (2014), and Koedel, Mihaly, and Rockoff (2015) for surveys). Estimating the effect of teacher characteristics on student performance is complicated because unobserved factors correlated with both teacher characteristics and student outcomes might bias estimates using non-experimental data. In this context, value-added models that focus on gains in student achievement help to eliminate the confounding impact of unobserved but fixed heterogeneity in student abilities across classes (Rockoff (2004); Rivkin, Hanushek, and Kain (2005); Aaronson, Barrow, and Sander (2007)). Very rich datasets covering multiple years of information for teachers even allow for estimating value-added models including a three-way fixed effects approach (student, teacher and school fixed effects). This provides the opportunity to link estimated teacher fixed effects with teacher characteristics or future outcomes of students (e.g., Chetty, Friedman, and Rockoff (2014); Cook and Mansfield (2016)). However, even in the most sophisticated models in this literature, extreme cases of non-random and dynamic sorting of teachers to students may give rise to remaining endogeneity concerns (e.g., Rothstein (2010)).

Fixed-effects panel approaches are not limited to the estimation of teacher effects or to exploiting individual-level treatments. For example, several studies on charter schools have employed student fixed effects approaches, which are not affected by the limitation of lottery studies of being restricted to oversubscribed schools (e.g., Imberman (2010), Booker, Sass, Gill, and Zimmer (2011), Baude, Casey, Hanushek, Phelan, and Rivkin (2018); see the survey by Epple et al. (2016)). Exploiting reforms of student assessment systems at the country level over time, Bergbauer, Hanushek, and Woessmann (2018) use country fixed effects to

take out unobserved time-invariant country heterogeneity when estimating the effects of testing reforms in a cross-country analysis.

Using the time dimension is only one approach to take out fixed effects. Even without having observations for the same unit at several points in time, one can exclude fixed effects by using special datasets. Specific studies are able to use information on contemporaneously observed academic achievements in two subjects of the same individual, which allows them to take out fixed effects of a student that affect all subjects alike. Studies that use this approach include Dee (2005) on teacher-student matches, Schwerdt and Wuppermann (2011) on teaching methods, Metzler and Woessmann (2012) on teacher subject knowledge, Lavy (2015) on instruction time, and Falck, Mang, and Woessmann (2018) on computer use. Other approaches use data on siblings or twins to address unobserved heterogeneity (e.g., Black, Devereux, and Salvanes (2007); Kalil, Mogstad, Rege, and Votruba (2016)).

However, despite the substantial advancements made over recent years in the literature, it remains a fact that panel-data techniques can account for unobserved heterogeneity only inasmuch as the relevant unobserved characteristics do not differ systematically over the panel dimension. Unobserved interferences that are not constant over time (or subjects or twins) cannot be fixed by standard panel-data techniques alone and require the experimental methods discussed above.

Conclusions

This article has described empirical techniques that researchers have designed to allow for causal inference by overcoming the problem that unobserved confounding factors may give rise to spurious associations. The techniques described here are, however, by no means the only ones used in the economics of education literature to address causal questions. Also, the insight that experimental and quasi-experimental studies are best positioned to discern causal effects does not invalidate studies using more traditional techniques such as standard multivariate regressions. But the methods discussed in this article caution about possible limits to the causal interpretation of estimates and help to point out how more traditional research designs can yield results that do not capture causal effects.

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